However, we can use their centroids to bootstrap k-means, replacing its default (random) initialization step with a more balanced choice.

Alternatively, we could run canopy clustering with coarse values of T_1 and T_2, and improve the initialization step by making sure to draw a fraction of the initial centroids from each of these areas: if the canopy clustering returned `m<=k` pseudo-clusters, draw `k/m` centroids from each of them. In practical experiments, this bootstrapping provided a relevant speed-up in convergence for k-means.

If you remember what we discussed in chapter 12, DBSCAN (section 12.3) has a weak spot when applied to datasets with non-uniform density, and OPTICS (section 12.4) can partially remedy to this issue, at the cost of a heavier computational load, and some experimenting with its parameters. Ideally, what we would need is being able to run DBSCAN independently on areas with different density, and tuning its parameters (or just the value for ε) for each of these areas separately.

Using canopy clustering as a first step can help us with this issue: if we run DBSCAN on each pseudo-cluster separately, we can expect smaller regions to have a more uniform density, and regions with different density to be – likely – assigned to different pre-clusters.